

data_wr

April 24, 2025

0.1 Data Extraction

```
[1]: # Import necessary libraries
import pandas as pd
import os

# Define function to extract specific columns
def extract_columns(input_file, output_file, columns_to_keep,
    ↪columns_rename={}):
    """
    Extract specific columns from a CSV file and save to a new CSV file.

    Parameters:
    input_file (str): Path to the input CSV file
    output_file (str): Path to save the output CSV file
    columns_to_keep (list): List of column names to keep
    columns_rename (dict): Dictionary mapping original column names to new names

    Returns:
    DataFrame: The extracted data
    """
    try:
        # Read the CSV file
        df = pd.read_csv(input_file)

        # Display the first few rows of the original dataframe
        print("First 5 rows of the original dataframe:")
        display(df.head())

        # Print all column names for reference
        print("\nAll columns in the CSV file:")
        print(df.columns.tolist())

        # Check if all specified columns exist in the dataframe
        missing_columns = [col for col in columns_to_keep if col not in df.
    ↪columns]
        if missing_columns:
```

```

        print(f"\nWarning: The following columns were not found in the CSV:␣
↪{missing_columns}")
        # Filter out missing columns
        columns_to_keep = [col for col in columns_to_keep if col in df.
↪columns]

        if not columns_to_keep:
            print("Error: None of the specified columns were found in the CSV␣
↪file.")
            return None

        # Select only the specified columns
        df_selected = df[columns_to_keep]

        # Rename columns if specified
        if columns_rename:
            # Filter out any rename mappings for columns that don't exist
            valid_renames = {old: new for old, new in columns_rename.items() if␣
↪old in df_selected.columns}
            if valid_renames:
                df_selected = df_selected.rename(columns=valid_renames)
                print("\nRenamed the following columns:")
                for old, new in valid_renames.items():
                    print(f"  {old} -> {new}")

            # Check for invalid rename mappings
            invalid_renames = {old: new for old, new in columns_rename.items()␣
↪if old not in df_selected.columns}
            if invalid_renames:
                print("\nWarning: Could not rename the following columns (not␣
↪found in selection):")
                for old, new in invalid_renames.items():
                    print(f"  {old} -> {new}")

        # Display the first few rows of the new dataframe
        print("\nFirst 5 rows of the extracted dataframe:")
        display(df_selected.head())

        # Save the selected columns to a new CSV file
        df_selected.to_csv(output_file, index=False)

        print(f"\nSuccessfully extracted {len(columns_to_keep)} columns to␣
↪{output_file}")
        return df_selected

    except Exception as e:

```

```

print(f"Error: {str(e)}")
return None

```

```

[2]: # Configuration cells - modify these values
input_file = "data/children_2015.csv" # Replace with your actual input file,
    ↪path
output_file = "exports/children_extr_2015.csv" # Replace with your desired,
    ↪output file path

# Replace with your identified column names
columns_to_keep = [
    "h22",
    # "ml13e",
    "v001",
    "v021",
    "v024",
    "v025",
    "b5",
    "b8",
    "m10",
]

column_rename_dict = {
    "h22": "treat_src",      # Source of treatment
    # "ml13e": "act_used",   # ACT use (Antimalarial)
    "v001": "cluster",      # Cluster number
    "v021": "stratum",      # Sample stratum
    "v024": "region",       # Region
    "v025": "residence",    # Urban/rural
    "b5": "alive",          # Child alive
    "b8": "age_mo",         # Age in months
    "m10": "fever_rec"      # Fever recall
}

# Execute the function in Jupyter
if not os.path.exists(input_file):
    print(f"Error: Input file '{input_file}' not found.")
else:
    result_df = extract_columns(input_file, output_file, columns_to_keep,
    ↪column_rename_dict)

```

First 5 rows of the original dataframe:

C:\Users\roywaswa\AppData\Local\Temp\ipykernel_28928\2086272929.py:21:

DtypeWarning: Columns (276) have mixed types. Specify dtype option on import or set low_memory=False.

```
df = pd.read_csv(input_file)
```

	caseid	hidx	v000	v001	v002	v003	v004	v005	v006	v007	\
0	001027	02	1 KE6	1	27	2	1	6074405	7	2015	
1	001027	02	2 KE6	1	27	2	1	6074405	7	2015	
2	001048	02	1 KE6	1	48	2	1	6074405	7	2015	
3	001048	02	2 KE6	1	48	2	1	6074405	7	2015	
4	001076	02	1 KE6	1	76	2	1	6074405	7	2015	

	sv112g	sv112h	sv112i	sv112j	sv112k	sv112l	sv112m	sv112n	sv112o	\
0	...	no	yes	yes	yes	no	no	no	no	yes
1	...	no	yes	yes	yes	no	no	no	no	yes
2	...	no	yes	yes	yes	yes	no	no	no	no
3	...	no	yes	yes	yes	yes	no	no	no	no
4	...	no	yes	yes	yes	yes	yes	no	no	yes

	sv112p
0	yes
1	yes
2	yes
3	yes
4	yes

[5 rows x 332 columns]

All columns in the CSV file:

['caseid', 'hidx', 'v000', 'v001', 'v002', 'v003', 'v004', 'v005', 'v006', 'v007', 'v008', 'v009', 'v010', 'v011', 'v012', 'v013', 'v014', 'v015', 'v016', 'v017', 'v018', 'v019', 'v019a', 'v020', 'v021', 'v022', 'v023', 'v024', 'v025', 'v026', 'v027', 'v028', 'v029', 'v030', 'v031', 'v032', 'v034', 'v040', 'v042', 'v044', 'v101', 'v102', 'v103', 'v104', 'v105', 'v106', 'v107', 'v113', 'v115', 'v116', 'v119', 'v120', 'v121', 'v122', 'v123', 'v124', 'v125', 'v127', 'v128', 'v129', 'v130', 'v131', 'v133', 'v134', 'v135', 'v136', 'v137', 'v138', 'v139', 'v140', 'v141', 'v149', 'v150', 'v151', 'v152', 'v153', 'awfactt', 'awfactu', 'awfactr', 'awfacte', 'awfactw', 'v155', 'v156', 'v157', 'v158', 'v159', 'v160', 'v161', 'v166', 'v167', 'v168', 'v190', 'v191', 'ml101', 'v201', 'v202', 'v203', 'v204', 'v205', 'v206', 'v207', 'v208', 'v209', 'v210', 'v211', 'v212', 'v213', 'v214', 'v215', 'v216', 'v217', 'v218', 'v219', 'v220', 'v221', 'v222', 'v223', 'v224', 'v225', 'v226', 'v227', 'v228', 'v229', 'v230', 'v231', 'v232', 'v233', 'v234', 'v235', 'v237', 'v238', 'v239', 'v240', 'v241', 'v242', 'v243', 'bidx', 'bord', 'b0', 'b1', 'b2', 'b3', 'b4', 'b5', 'b6', 'b7', 'b8', 'b9', 'b10', 'b11', 'b12', 'b13', 'b15', 'b16', 's216d', 'midx', 'm2a', 'm2b', 'm2c', 'm2d', 'm2e', 'm2f', 'm2g', 'm2h', 'm2i', 'm2j', 'm2k', 'm2l', 'm2m', 'm2n', 'm14', 'm49a', 'h22', 'h31', 'h31b', 'h31c', 'h31d', 'h31e', 'h32a', 'h32b', 'h32c', 'h32d', 'h32e', 'h32f', 'h32g', 'h32h', 'h32i', 'h32j', 'h32k', 'h32l', 'h32m', 'h32n', 'h32o', 'h32p', 'h32q', 'h32r', 'h32s', 'h32t', 'h32u', 'h32v', 'h32w', 'h32x', 'h32y', 'h32z', 'h37a', 'h37b', 'h37c', 'h37d', 'h37e', 'h37f', 'h37g', 'h37h', 'h37i', 'h37j', 'h37k', 'h37l', 'h37m', 'h37n', 'h37o', 'h37p', 'h37x', 'h37y', 'h37z', 'h46a', 'h46b', 'h47', 'idxml', 'ml0', 'ml1', 'ml2', 'ml11',

```
'ml12', 'ml13a', 'ml13b', 'ml13c', 'ml13d', 'ml13e', 'ml13f', 'ml13g', 'ml13h',
'ml13i', 'ml13j', 'ml13k', 'ml13l', 'ml13m', 'ml13n', 'ml13o', 'ml13p', 'ml13x',
'ml13y', 'ml13z', 'ml14a', 'ml14b', 'ml14y', 'ml14z', 'ml15a', 'ml15b', 'ml15c',
'ml16a', 'ml16b', 'ml16c', 'ml17a', 'ml17b', 'ml17c', 'ml18a', 'ml18b', 'ml18c',
'ml19a', 'ml19b', 'ml19c', 'ml19d', 'ml19e', 'ml19f', 'ml19x', 'ml19y', 'ml19z',
'ml20a', 'ml20b', 'ml20c', 'ml21a', 'ml21b', 'ml21c', 'ml22a', 'ml22b', 'ml22c',
'ml23a', 'ml23b', 'ml23c', 'ml24c', 's411a', 's411b', 's411c', 'ml25a', 's412b',
'scounty', 'szone', 'slang', 'slangint', 'stransl', 's105', 's106', 's111',
's112a', 's112b', 's112c', 's112d', 's112e', 's112f', 's427b', 's427c', 's427d',
's427e', 's427f1', 's427f2', 's427f3', 's427f4', 's427f5', 's427f6', 's427f7',
's427f8', 's427f9', 'sv112e', 'sv112g', 'sv112h', 'sv112i', 'sv112j', 'sv112k',
'sv112l', 'sv112m', 'sv112n', 'sv112o', 'sv112p']
```

Renamed the following columns:

```
h22 -> treat_src
v001 -> cluster
v021 -> stratum
v024 -> region
v025 -> residence
b5 -> alive
b8 -> age_mo
ml0 -> fever_rec
```

Warning: Could not rename the following columns (not found in selection):

```
h22 -> treat_src
v001 -> cluster
v021 -> stratum
v024 -> region
v025 -> residence
b5 -> alive
b8 -> age_mo
ml0 -> fever_rec
```

First 5 rows of the extracted dataframe:

	treat_src	cluster	stratum	region	residence	alive	age_mo	\
0	no	1	1	nairobi	urban	yes	2.0	
1	no	1	1	nairobi	urban	yes	3.0	
2	yes	1	1	nairobi	urban	yes	1.0	
3	no	1	1	nairobi	urban	yes	3.0	
4	yes	1	1	nairobi	urban	yes	4.0	

	fever_rec
0	only treated nets
1	only treated nets
2	only treated nets
3	only treated nets
4	NaN

Successfully extracted 8 columns to exports/children_extr_2015.csv

```
[3]: # Configuration cells - modify these values
input_file = "data/children_2020.csv" # Replace with your actual input file,
    ↪path
output_file = "exports/children_extr_2020.csv" # Replace with your desired,
    ↪output file path

# Replace with your identified column names
columns_to_keep = [
    "h22",
    "m10",
    "b5",
    "b8",
    # "ml13aa",
    # "ml13ab",
    "v001",
    "v021",
    "v024",
    "v025"
]

column_names = {
    "h22": "treat_src",
    "m10": "fever_rec",
    "b5": "alive",
    "b8": "age_mo",
    # "ml13aa": "net1",
    # "ml13ab": "net2",
    "v001": "cluster",
    "v021": "stratum",
    "v024": "region",
    "v025": "residence"
}

# Execute the function in Jupyter
if not os.path.exists(input_file):
    print(f"Error: Input file '{input_file}' not found.")
else:
    result_df = extract_columns(input_file, output_file, columns_to_keep,
    ↪column_names)
```

First 5 rows of the original dataframe:

caseid	hidxa	v000	v001	v002	v003	v004	v005	v006	v007	\
--------	-------	------	------	------	------	------	------	------	------	---

0	2	1	2	1	KE7	2	1	2	2	607390	12	2020
1	2	1	2	2	KE7	2	1	2	2	607390	12	2020
2	2	6	2	1	KE7	2	6	2	2	607390	11	2020
3	2	6	2	2	KE7	2	6	2	2	607390	11	2020
4	2	10	2	1	KE7	2	10	2	2	607390	11	2020

	...	s307a	s309ba	s309bb	s309bc	s309bd	s309be	s309bf	s309bg	\
0	...	spouse	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
2	...	respondent	no	no	no	no	no	no	no	
3	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
4	...	respondent	no	no	no	no	no	yes	no	

	s309bx	s309bz
0	NaN	NaN
1	NaN	NaN
2	yes	no
3	NaN	NaN
4	no	no

[5 rows x 383 columns]

All columns in the CSV file:

['caseid', 'hidxa', 'v000', 'v001', 'v002', 'v003', 'v004', 'v005', 'v006', 'v007', 'v008', 'v008a', 'v009', 'v010', 'v011', 'v012', 'v013', 'v014', 'v015', 'v016', 'v021', 'v022', 'v023', 'v024', 'v025', 'v026', 'v027', 'v028', 'v029', 'v030', 'v031', 'v032', 'v034', 'v040', 'v042', 'v045a', 'v045b', 'v045c', 'v046', 'v101', 'v102', 'v106', 'v107', 'v113', 'v115', 'v116', 'v119', 'v120', 'v121', 'v122', 'v123', 'v124', 'v125', 'v127', 'v128', 'v129', 'v130', 'v131', 'v133', 'v134', 'v135', 'v136', 'v137', 'v138', 'v139', 'v140', 'v141', 'v149', 'v150', 'v151', 'v152', 'v153', 'v155', 'v157', 'v158', 'v159', 'v160', 'v161', 'v169a', 'v171a', 'v171b', 'v190', 'v191', 'v190a', 'v191a', 'ml101', 'v201', 'v202', 'v203', 'v204', 'v205', 'v206', 'v207', 'v208', 'v209', 'v210', 'v213', 'v214', 'v218', 'v219', 'v220', 'v222', 'v223', 'v224', 'v237', 'v238', 'v417', 'v418a', 'v419', 'v459', 'v460', 'v461', 'v468', 'bidx', 'bord', 'b0', 'b1', 'b2', 'b3', 'b4', 'b5', 'b6', 'b7', 'b8', 'b9', 'b10', 'b11', 'b12', 'b13', 'b15', 'b16', 'b17', 'b18', 'b19', 'midx', 'm2a', 'm2b', 'm2c', 'm2d', 'm2e', 'm2f', 'm2g', 'm2h', 'm2i', 'm2j', 'm2k', 'm2l', 'm2m', 'm2n', 'm13', 'm14', 'm49a', 'm49b', 'm49c', 'm49d', 'm49e', 'm49f', 'm49g', 'm49x', 'm49z', 'm49y', 'm57a', 'm57b', 'm57c', 'm57d', 'm57e', 'm57f', 'm57g', 'm57h', 'm57i', 'm57j', 'm57k', 'm57l', 'm57m', 'm57n', 'm57o', 'm57p', 'm57q', 'm57r', 'm57s', 'm57t', 'm57u', 'm57v', 'm57x', 'h22', 'h31', 'h32a', 'h32b', 'h32c', 'h32d', 'h32e', 'h32f', 'h32g', 'h32h', 'h32i', 'h32j', 'h32k', 'h32l', 'h32m', 'h32n', 'h32o', 'h32p', 'h32q', 'h32r', 'h32s', 'h32t', 'h32u', 'h32v', 'h32w', 'h32x', 'h32y', 'h32z', 'h46a', 'h46b', 'h47', 's406', 'hwidx', 'hw1', 'hw16', 'hw17', 'hw18', 'hw19', 'hw51', 'hw52', 'hw53', 'hw55', 'hw56', 'hw57', 'hw58', 'v801', 'v802', 'v803', 'v804', 'v805', 'v806', 'idxml', 'ml0', 'ml1', 'ml2', 'ml11', 'ml12',

```
'ml13a', 'ml13b', 'ml13c', 'ml13d', 'ml13da', 'ml13e', 'ml13aa', 'ml13ab',
'ml13f', 'ml13g', 'ml13h', 'ml13i', 'ml13j', 'ml13k', 'ml13l', 'ml13m', 'ml13n',
'ml13o', 'ml13p', 'ml13x', 'ml13y', 'ml13z', 'ml14a', 'ml14b', 'ml14y', 'ml14z',
'ml15a', 'ml15b', 'ml15c', 'ml16a', 'ml16b', 'ml16c', 'ml17a', 'ml17b', 'ml17c',
'ml18a', 'ml18b', 'ml18c', 'ml19a', 'ml19b', 'ml19c', 'ml19d', 'ml19e', 'ml19f',
'ml19x', 'ml19y', 'ml19z', 'ml20a', 'ml20b', 'ml20c', 'ml21a', 'ml21b', 'ml21c',
'ml22a', 'ml22b', 'ml22c', 'ml23a', 'ml23b', 'ml23c', 'ml24c', 'ml25a', 's413a',
's413b', 's413c', 'scounty', 'szone', 's114', 's211', 's225', 's501', 's502a',
's502b', 's502c', 's502d', 's502e', 's502f', 's502g', 's502h', 's502i', 's502j',
's502k', 's502l', 's502m', 's502x', 's502z', 's502aa', 's502ab', 's502ac',
's502ad', 's502ae', 's502af', 's502ag', 's502ax', 's502az', 's503', 's504a',
's504b', 's504c', 's504d', 's504e', 's504f', 's504g', 's504h', 's504i', 's504x',
's504z', 's504aa', 's504ab', 's504ac', 's504ad', 's504ax', 's504az', 's505',
's506', 's507', 's508', 's509', 's510', 's511', 's511a', 's512', 's513', 's514',
's514a', 's514ba', 's514bb', 's514bc', 's514bd', 's514bx', 's514bz', 'idx94',
's306', 's307a', 's309ba', 's309bb', 's309bc', 's309bd', 's309be', 's309bf',
's309bg', 's309bx', 's309bz']
```

Renamed the following columns:

```
h22 -> treat_src
ml0 -> fever_rec
b5 -> alive
b8 -> age_mo
v001 -> cluster
v021 -> stratum
v024 -> region
v025 -> residence
```

Warning: Could not rename the following columns (not found in selection):

```
h22 -> treat_src
ml0 -> fever_rec
b5 -> alive
b8 -> age_mo
v001 -> cluster
v021 -> stratum
v024 -> region
v025 -> residence
```

First 5 rows of the extracted dataframe:

	treat_src	fever_rec	alive	age_mo	cluster	stratum	region \
0	yes	no net	yes	0.0	2	2	coast
1	no	only treated nets	yes	4.0	2	2	coast
2	no	no net	yes	2.0	2	2	coast
3	no	no net	yes	3.0	2	2	coast
4	yes	only treated nets	yes	0.0	2	2	coast

residence


```

0    rural
1    rural
2    rural
3    rural
4    rural

```

Successfully extracted 8 columns to exports/children_extr_2020.csv

```

[4]: # Configuration cells - modify these values
input_file = "data/household_member-2015.csv" # Replace with your actual input_
      ↳file path
output_file = "exports/household_member-extr_2015.csv" # Replace with your_
      ↳desired output file path

# Replace with your identified column names
columns_to_keep = [
    "hml12",
    "hml20",
    "hv001",
    "hv024",
    "hv021",
    "hv025",
]

column_rename_dict = {
    "hml12": "net_owned",      # Number of mosquito nets owned
    "hml20": "net_used",      # Used net last night
    "hv001": "cluster",       # Cluster number
    "hv024": "region",        # Region
    "hv021": "stratum",       # Sample stratum
    "hv025": "residence"      # Urban/rural
}

# Execute the function in Jupyter
if not os.path.exists(input_file):
    print(f"Error: Input file '{input_file}' not found.")
else:
    result_df = extract_columns(input_file, output_file, columns_to_keep,
    ↳column_rename_dict)

```

C:\Users\roywaswa\AppData\Local\Temp\ipykernel_28928\2086272929.py:21:
DtypeWarning: Columns (137,140,167,182,183,184,185,186,187,188,189,190,191,192,193,196,197,198,199,200,201,264,325,338,339,340,341,342,343,344,345,346,347) have mixed types. Specify dtype option on import or set low_memory=False.

```
df = pd.read_csv(input_file)
```

First 5 rows of the original dataframe:

	hhid	hvidx	hv000	hv001	hv002	hv003	hv004	hv005	hv006	hv007	...	\
0	1002	1	KE6	1	2	1	1	6021054	7	2015	...	
1	1006	1	KE6	1	6	1	1	6021054	7	2015	...	
2	1009	1	KE6	1	9	1	1	6021054	7	2015	...	
3	1013	1	KE6	1	13	1	1	6021054	7	2015	...	
4	1013	2	KE6	1	13	1	1	6021054	7	2015	...	

	shb118d	shb118e	shb118f	shb118g	shb118h	shb121	shb125	shml16a	\
0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
2	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
3	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
4	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	

	shb108t	shb110t
0	NaN	NaN
1	NaN	NaN
2	NaN	NaN
3	NaN	NaN
4	NaN	NaN

[5 rows x 351 columns]

All columns in the CSV file:

```
['hhid', 'hvidx', 'hv000', 'hv001', 'hv002', 'hv003', 'hv004', 'hv005', 'hv006',
'hv007', 'hv008', 'hv009', 'hv010', 'hv011', 'hv012', 'hv013', 'hv014', 'hv015',
'hv016', 'hv017', 'hv018', 'hv019', 'hv020', 'hv021', 'hv022', 'hv023', 'hv024',
'hv025', 'hv026', 'hv027', 'hv028', 'hv030', 'hv031', 'hv032', 'hv035', 'hv040',
'hv041', 'hv042', 'hv044', 'hv201', 'hv202', 'hv204', 'hv205', 'hv206', 'hv207',
'hv208', 'hv209', 'hv210', 'hv211', 'hv212', 'hv213', 'hv214', 'hv215', 'hv216',
'hv217', 'hv218', 'hv219', 'hv220', 'hv221', 'hv225', 'hv226', 'hv227', 'hv228',
'hv230a', 'hv230b', 'hv232', 'hv232b', 'hv232c', 'hv232d', 'hv232e', 'hv232y',
'hv234', 'hv234a', 'hv235', 'hv236', 'hv237', 'hv237a', 'hv237b', 'hv237c',
'hv237d', 'hv237e', 'hv237f', 'hv237g', 'hv237h', 'hv237i', 'hv237j', 'hv237k',
'hv237x', 'hv237z', 'hv238', 'hv239', 'hv240', 'hv241', 'hv242', 'hv243a',
'hv243b', 'hv243c', 'hv243d', 'hv244', 'hv245', 'hv246', 'hv246a', 'hv246b',
'hv246c', 'hv246d', 'hv246e', 'hv246f', 'hv246g', 'hv246h', 'hv246i', 'hv246j',
'hv246k', 'hv247', 'hv252', 'hv253', 'hv253a', 'hv253b', 'hv253c', 'hv253d',
'hv253e', 'hv253f', 'hv253g', 'hv253h', 'hv253x', 'hv253z', 'hv270', 'hv271',
'hml1', 'hml1a', 'hml2', 'shcounty', 'shzone', 'shbegin', 'shend', 'shlang',
'shlangint', 'shtransl', 'sh101a', 'sh109b', 'sh111u', 'sh111', 'sh112e',
'sh112g', 'sh112h', 'sh112i', 'sh112j', 'sh112k', 'sh112l', 'sh112m', 'sh112n',
'sh112o', 'sh112p', 'sh113aa', 'sh113bb', 'sh113cc', 'sh128a', 'sh128b',
'sh128c', 'sh128d', 'sh128e1', 'sh128e2', 'sh128e3', 'sh128e4', 'sh128e5',
```

```

'sh128e6', 'sh128e7', 'sh128e8', 'sh128f', 'sh128g1', 'sh128g2', 'sh128g3',
'sh128g4', 'sh128g5', 'sh128g6', 'sh128g7', 'sh128g8', 'sh128g9', 'sh128g10',
'sh128h1', 'sh128h2', 'sh128h3', 'sh128h4', 'sh128i', 'sh128j', 'sh128k',
'sh128l', 'sh128m', 'sh128n', 'sh128o', 'sh128p', 'sh128q', 'sh128r', 'sh128s',
'sh128t', 'sh128u', 'sh128v', 'sh128x1', 'sh128x2', 'sh128x3', 'sh128x4',
'sh128x5', 'sh128y', 'sh128z', 'sh129a', 'sh129b', 'sh129c', 'sh129d', 'sh129e',
'sh129f', 'sh129g', 'sh129h', 'hv101', 'hv102', 'hv103', 'hv104', 'hv105',
'hv106', 'hv107', 'hv108', 'hv109', 'hv110', 'hv111', 'hv112', 'hv113', 'hv114',
'hv115', 'hv116', 'hv117', 'hv118', 'hv120', 'hv121', 'hv122', 'hv123', 'hv124',
'hv125', 'hv126', 'hv127', 'hv128', 'hv129', 'hv140', 'hc0', 'hc1', 'hc2',
'hc3', 'hc4', 'hc5', 'hc6', 'hc7', 'hc8', 'hc9', 'hc10', 'hc11', 'hc12', 'hc13',
'hc15', 'hc16', 'hc17', 'hc18', 'hc19', 'hc27', 'hc30', 'hc31', 'hc32', 'hc33',
'hc51', 'hc52', 'hc53', 'hc55', 'hc56', 'hc57', 'hc58', 'hc60', 'hc61', 'hc62',
'hc63', 'hc64', 'hc68', 'hc70', 'hc71', 'hc72', 'hc73', 'shc1', 'hmlidx',
'hml3', 'hml4', 'hml5', 'hml6', 'hml7', 'hml8', 'hml9', 'hml10', 'hml11',
'hml21', 'hmla', 'hmlb', 'hmlc', 'hmdl', 'hmle', 'sh121a', 'sh125', 'sh125a',
'sh126a1', 'sh126a2', 'sh126a3', 'sh126a4', 'sh126a5', 'sh126a6', 'sh126a7',
'sh127bb', 'sh127cc1', 'sh127cc2', 'sh127cc3', 'sh127cc4', 'sh127cc5', 'hmhidx',
'hml12', 'hml13', 'hml14', 'hml15', 'hml16', 'hml16a', 'hml17', 'hml18',
'hml19', 'hml20', 'hml30', 'hml31', 'hml32', 'hml32a', 'hml32b', 'hml32c',
'hml32d', 'hml32e', 'hml32f', 'hml32g', 'hml33', 'hml34', 'hml35', 'shb118a',
'shb118b', 'shb118c', 'shb118d', 'shb118e', 'shb118f', 'shb118g', 'shb118h',
'shb121', 'shb125', 'shml16a', 'shb108t', 'shb110t']

```

Renamed the following columns:

```

hml12 -> net_owned
hml20 -> net_used
hv001 -> cluster
hv024 -> region
hv021 -> stratum
hv025 -> residence

```

Warning: Could not rename the following columns (not found in selection):

```

hml12 -> net_owned
hml20 -> net_used
hv001 -> cluster
hv024 -> region
hv021 -> stratum
hv025 -> residence

```

First 5 rows of the extracted dataframe:

		net_owned	net_used	cluster	region	stratum	residence
0	only treated (itn) nets		yes	1	nairobi	1	urban
1	did not sleep under a net		no	1	nairobi	1	urban
2	only treated (itn) nets		yes	1	nairobi	1	urban
3	only treated (itn) nets		yes	1	nairobi	1	urban
4	only treated (itn) nets		yes	1	nairobi	1	urban

Successfully extracted 6 columns to exports/household_member-extr_2015.csv

```
[5]: # Configuration cells - modify these values
input_file = "data/household_member-2020.csv" # Replace with your actual input_
      ↪file path
output_file = "exports/household_member-extr_2020.csv" # Replace with your_
      ↪desired output file path

# Replace with your identified column names
columns_to_keep = [
    "hml12",
    "shcounty",
    "hv001",
    "hv024",
    "hv021",
    "hv025",
]

column_rename_dict = {
    "hml12": "net_owned", # Number of ITNs in the household
    "shcounty": "county", # County name (custom DHS variable)
    "hv001": "cluster", # Cluster number
    "hv024": "region", # Region/province
    "hv021": "stratum", # Sampling stratum
    "hv025": "residence" # Urban/rural residence
}

# Execute the function in Jupyter
if not os.path.exists(input_file):
    print(f"Error: Input file '{input_file}' not found.")
else:
    result_df = extract_columns(input_file, output_file, columns_to_keep,
    ↪column_rename_dict)
```

C:\Users\roywaswa\AppData\Local\Temp\ipykernel_28928\2086272929.py:21:

DtypeWarning: Columns (53,90) have mixed types. Specify dtype option on import or set low_memory=False.

```
df = pd.read_csv(input_file)
```

First 5 rows of the original dataframe:

	hhid	hvidx	hv000	hv001	hv002	hv003	hv004	hv005	hv006	\
0	2	1	1	KE7	2	1	8	2	588318	11
1	2	1	2	KE7	2	1	8	2	588318	11

2	2	1	3	KE7	2	1	8	2	588318	11
3	2	1	4	KE7	2	1	8	2	588318	11
4	2	1	5	KE7	2	1	8	2	588318	11

	hv007	...	hml32c	hml32d	hml32e	hml32f	hml32g	hml33	hml34	\
0	2020	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1	2020	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
2	2020	...	no	no	NaN	NaN	NaN	measured	C5B7P	
3	2020	...	no	no	NaN	NaN	NaN	measured	P6G7J	
4	2020	...	no	no	NaN	NaN	NaN	measured	B0V4B	

	hml35	hml36	sml16a
0	NaN	NaN	NaN
1	NaN	NaN	NaN
2	negative	305.0	138.0
3	negative	305.0	114.0
4	negative	305.0	80.0

[5 rows x 223 columns]

All columns in the CSV file:

['hhid', 'hvidx', 'hv000', 'hv001', 'hv002', 'hv003', 'hv004', 'hv005', 'hv006', 'hv007', 'hv008', 'hv008a', 'hv009', 'hv010', 'hv011', 'hv012', 'hv013', 'hv014', 'hv015', 'hv016', 'hv017', 'hv018', 'hv019', 'hv020', 'hv021', 'hv022', 'hv023', 'hv024', 'hv025', 'hv026', 'hv027', 'hv028', 'hv030', 'hv031', 'hv032', 'hv035', 'hv040', 'hv041', 'hv042', 'hv045a', 'hv045b', 'hv045c', 'hv046', 'hv801', 'hv802', 'hv803', 'hv804', 'hv807d', 'hv807m', 'hv807y', 'hv807c', 'hv807a', 'hv201', 'hv202', 'hv201a', 'hv204', 'hv205', 'hv206', 'hv207', 'hv208', 'hv209', 'hv210', 'hv211', 'hv212', 'hv213', 'hv214', 'hv215', 'hv216', 'hv217', 'hv218', 'hv219', 'hv220', 'hv221', 'hv225', 'hv226', 'hv227', 'hv228', 'hv235', 'hv238', 'hv238a', 'hv239', 'hv240', 'hv241', 'hv242', 'hv243a', 'hv243b', 'hv243c', 'hv243d', 'hv243e', 'hv244', 'hv245', 'hv246', 'hv246a', 'hv246b', 'hv246c', 'hv246d', 'hv246e', 'hv246f', 'hv246g', 'hv246h', 'hv246i', 'hv246j', 'hv246k', 'hv247', 'hv270', 'hv271', 'hv270a', 'hv271a', 'hml1', 'hml1a', 'hml2', 'shcounty', 'shzone', 'sh109', 'sh110', 'sh116g', 'sh116h', 'sh116i', 'sh116j', 'sh116k', 'sh116l', 'sh116m', 'sh116n', 'sh116o', 'sh116p', 'sh119', 'sh119a', 'hv101', 'hv102', 'hv103', 'hv104', 'hv105', 'hv117', 'hv120', 'hc0', 'hc1', 'hc1a', 'hc16', 'hc17', 'hc18', 'hc19', 'hc20', 'hc21', 'hc27', 'hc30', 'hc31', 'hc32', 'hc32a', 'hc33', 'hc51', 'hc52', 'hc53', 'hc55', 'hc56', 'hc57', 'hc58', 'hc60', 'hc61', 'hc62', 'hc63', 'hc64', 'hc68', 'shc1', 'shc1a', 'sb115a', 'sb115b', 'sb115c', 'sb115d', 'sb115e', 'sb115f', 'sb115g', 'sb115h', 'sb119', 'sb122', 'sb123', 'sb128d', 'sb128m', 'sb128y', 'hmlidx', 'hml3', 'hml4', 'hml5', 'hml6', 'hml7', 'hml8', 'hml9', 'hml10', 'hml11', 'hml21', 'hmla', 'hmlb', 'hmlc', 'hmld', 'hmle', 'hml22', 'hml23', 'sh130', 'hmhidx', 'hml12', 'hml13', 'hml14', 'hml15', 'hml16', 'hml16a', 'hml17', 'hml18', 'hml19', 'hml20', 'hml30', 'hml31', 'hml32', 'hml32a', 'hml32b', 'hml32c', 'hml32d', 'hml32e', 'hml32f', 'hml32g', 'hml33', 'hml34', 'hml35',

```
'hml36', 'sml16a']
```

Renamed the following columns:

```
hml12 -> net_owned
shcounty -> county
hv001 -> cluster
hv024 -> region
hv021 -> stratum
hv025 -> residence
```

Warning: Could not rename the following columns (not found in selection):

```
hml12 -> net_owned
shcounty -> county
hv001 -> cluster
hv024 -> region
hv021 -> stratum
hv025 -> residence
```

First 5 rows of the extracted dataframe:

		net_owned	county	cluster	region	stratum	residence
0	only treated (itn)	nets	kwale	2	coast	2	rural
1	only treated (itn)	nets	kwale	2	coast	2	rural
2	only treated (itn)	nets	kwale	2	coast	2	rural
3	only treated (itn)	nets	kwale	2	coast	2	rural
4	only treated (itn)	nets	kwale	2	coast	2	rural

Successfully extracted 6 columns to exports/household_member-extr_2020.csv

```
[6]: # Configuration cells - modify these values
input_file = "data/household_2020.csv" # Replace with your actual input file
      ↪path
output_file = "exports/household_extr_2020.csv" # Replace with your desired
      ↪output file path

# Replace with your identified column names
columns_to_keep = [
    "hv001",
    "hv227",
    "hv012",
    "hmlidx_1",
    "hmlidx_2",
    "hmlidx_3",
    "hmlidx_4",
    "hmlidx_5",
    "hmlidx_6",
    "hmlidx_7"
```

```

]

column_rename_dict = {
    "hv001": "cluster",      # Cluster number
    "hv227": "net_use",      # Slept under an ITN last night
    "hv012": "hh_size",      # Household size
    "hmlidx_1": "net_idx1",  # ITN index 1 (person)
    "hmlidx_2": "net_idx2",  # ITN index 2 (person)
    "hmlidx_3": "net_idx3",  # ITN index 3 (person)
    "hmlidx_4": "net_idx4",  # ITN index 4 (person)
    "hmlidx_5": "net_idx5",  # ITN index 5 (person)
    "hmlidx_6": "net_idx6",  # ITN index 6 (person)
    "hmlidx_7": "net_idx7"   # ITN index 7 (person)
}

# Execute the function in Jupyter
if not os.path.exists(input_file):
    print(f"Error: Input file '{input_file}' not found.")
else:
    result_df = extract_columns(input_file, output_file, columns_to_keep,
                                ↪column_rename_dict)

```

C:\Users\roywaswa\AppData\Local\Temp\ipykernel_28928\2086272929.py:21:

DtypeWarning: Columns (52,70,89,162,163,164,165,166,167,168,169,170,171,172,173,186,187,188,189,190,191,192,193,194,195,196,197,210,211,212,213,214,215,216,217,218,219,220,221,234,235,236,237,238,239,240,241,242,243,244,245,246,247,248,249,250,251,252,253,254,255,256,257,258,266,267,268,282,283,284,285,286,287,288,289,290,291,292,293,306,307,308,309,310,311,312,313,314,315,316,317,344,345,346,347,348,350,351,352,364,365,366,367,368,370,505,506,507,508,509,510,511,512,605,606,607,608,609,610,611,612,618,619,620,621,622,623,624,625,626,645,646,647,648,649,650,651,652,661,662,663,664,665,667,685,686,687,688,689,690,691,692,725,726,727,728,729,730,731,732,764,765,766,767,768,769,770,771,772,785,786,787,788,789,790,865,866,867,868,869,870,920,922,923,924,925,926,927,928,940,942,943,944,945,946,947,948,960,962,963,964,965,966,967,968,980,982,983,984,985,986,987,988,1000,1002,1003,1004,1005,1006,1007,1008,1020,1022,1023,1024,1025,1026,1027,1028,1040,1042,1043,1044,1045,1046,1047,1048,1060,1062,1063,1064,1065,1066,1067,1068,1080,1082,1083,1084,1085,1086,1087,1088,1098,1100,1102,1103,1104,1105,1106,1107,1108,1165,1166,1167,1168,1169,1170,1171,1172,1217,1218,1321,1322,1323,1329,1330,1367,1368,1369,1370,1371,1372,1373,1374,1375,1376,1377,1378,1391,1392,1393,1394,1395,1396,1397,1398,1399,1400,1401,1451,1452,1453,1454,1455,1456,1457,1460,1461,1463,1471,1472,1473,1485,1486,1487,1488,1489,1490,1491,1492,1493,1494,1511,1512,1513,1514,1515,1516,1517,1518,1519,1520,1521,1522,1530,1531,1532,1533,1534,1535,1536,1537,1539,1541,1543,1559,1560,1561,1562,1563,1564,1565,1566,1567,1568,1569,1570,1583,1584,1585,1586,1587,1588,1589,1590,1591,1592,1593,1594,1596,1597,1598,1599,1600,1601,1602,1603,1604,1605,1606,1612,1631,1632,1633,1634,1635,1636,1637,1638,1639,1642,1655,1656,1657,1658,1659,1660,1661,1662,1679,1680,1681,1682,1683,1684,1685,1686,1687,1690,1703,1704,1705,1706,1707,1708,1709,1710,1711,1714,1727,1728,1729)

9,1730,1731,1732,1733,1734,1735,1738,1751,1752,1753,1754,1755,1756,1757,1758,1759,1762,1847,1848,1849,1850,1851,1852,1853,1854,1855,1858,1871,1872,1873,1874,1875,1876,1877,1878,1895,1896,1897,1898,1899,1900,1901,1902) have mixed types.

Specify dtype option on import or set low_memory=False.

```
df = pd.read_csv(input_file)
```

First 5 rows of the original dataframe:

	hhid	hv000	hv001	hv002	hv003	hv004	hv005	hv006	hv007	\
0	2	1	KE7	2	1	8	2	588318	11	2020
1	2	6	KE7	2	6	2	2	588318	11	2020
2	2	10	KE7	2	10	2	2	588318	11	2020
3	2	13	KE7	2	13	1	2	588318	11	2020
4	2	16	KE7	2	16	3	2	588318	11	2020

	hv008	...	sml16a_15	sml16a_16	sml16a_17	sml16a_18	sml16a_19	\
0	1451	...	NaN	NaN	NaN	NaN	NaN	
1	1451	...	NaN	NaN	NaN	NaN	NaN	
2	1451	...	NaN	NaN	NaN	NaN	NaN	
3	1451	...	NaN	NaN	NaN	NaN	NaN	
4	1451	...	NaN	NaN	NaN	NaN	NaN	

	sml16a_20	sml16a_21	sml16a_22	sml16a_23	sml16a_24
0	NaN	NaN	NaN	NaN	NaN
1	NaN	NaN	NaN	NaN	NaN
2	NaN	NaN	NaN	NaN	NaN
3	NaN	NaN	NaN	NaN	NaN
4	NaN	NaN	NaN	NaN	NaN

[5 rows x 1955 columns]

All columns in the CSV file:

```
['hhid', 'hv000', 'hv001', 'hv002', 'hv003', 'hv004', 'hv005', 'hv006', 'hv007',
'hv008', 'hv008a', 'hv009', 'hv010', 'hv011', 'hv012', 'hv013', 'hv014',
'hv015', 'hv016', 'hv017', 'hv018', 'hv019', 'hv020', 'hv021', 'hv022', 'hv023',
'hv024', 'hv025', 'hv026', 'hv027', 'hv028', 'hv030', 'hv031', 'hv032', 'hv035',
'hv040', 'hv041', 'hv042', 'hv045a', 'hv045b', 'hv045c', 'hv046', 'hv801',
'hv802', 'hv803', 'hv804', 'hv807d', 'hv807m', 'hv807y', 'hv807c', 'hv807a',
'hv201', 'hv202', 'hv201a', 'hv204', 'hv205', 'hv206', 'hv207', 'hv208',
'hv209', 'hv210', 'hv211', 'hv212', 'hv213', 'hv214', 'hv215', 'hv216', 'hv217',
'hv218', 'hv219', 'hv220', 'hv221', 'hv225', 'hv226', 'hv227', 'hv228', 'hv235',
'hv238', 'hv238a', 'hv239', 'hv240', 'hv241', 'hv242', 'hv243a', 'hv243b',
'hv243c', 'hv243d', 'hv243e', 'hv244', 'hv245', 'hv246', 'hv246a', 'hv246b',
'hv246c', 'hv246d', 'hv246e', 'hv246f', 'hv246g', 'hv246h', 'hv246i', 'hv246j',
'hv246k', 'hv247', 'hv270', 'hv271', 'hv270a', 'hv271a', 'hml1', 'hml1a',
'hml2', 'shcounty', 'shzone', 'sh109', 'sh110', 'sh116g', 'sh116h', 'sh116i',
'sh116j', 'sh116k', 'sh116l', 'sh116m', 'sh116n', 'sh116o', 'sh116p', 'sh119',
'sh119a', 'hvidx_01', 'hvidx_02', 'hvidx_03', 'hvidx_04', 'hvidx_05',
```


'hvidx_06', 'hvidx_07', 'hvidx_08', 'hvidx_09', 'hvidx_10', 'hvidx_11',
'hvidx_12', 'hvidx_13', 'hvidx_14', 'hvidx_15', 'hvidx_16', 'hvidx_17',
'hvidx_18', 'hvidx_19', 'hvidx_20', 'hvidx_21', 'hvidx_22', 'hvidx_23',
'hvidx_24', 'hv101_01', 'hv101_02', 'hv101_03', 'hv101_04', 'hv101_05',
'hv101_06', 'hv101_07', 'hv101_08', 'hv101_09', 'hv101_10', 'hv101_11',
'hv101_12', 'hv101_13', 'hv101_14', 'hv101_15', 'hv101_16', 'hv101_17',
'hv101_18', 'hv101_19', 'hv101_20', 'hv101_21', 'hv101_22', 'hv101_23',
'hv101_24', 'hv102_01', 'hv102_02', 'hv102_03', 'hv102_04', 'hv102_05',
'hv102_06', 'hv102_07', 'hv102_08', 'hv102_09', 'hv102_10', 'hv102_11',
'hv102_12', 'hv102_13', 'hv102_14', 'hv102_15', 'hv102_16', 'hv102_17',
'hv102_18', 'hv102_19', 'hv102_20', 'hv102_21', 'hv102_22', 'hv102_23',
'hv102_24', 'hv103_01', 'hv103_02', 'hv103_03', 'hv103_04', 'hv103_05',
'hv103_06', 'hv103_07', 'hv103_08', 'hv103_09', 'hv103_10', 'hv103_11',
'hv103_12', 'hv103_13', 'hv103_14', 'hv103_15', 'hv103_16', 'hv103_17',
'hv103_18', 'hv103_19', 'hv103_20', 'hv103_21', 'hv103_22', 'hv103_23',
'hv103_24', 'hv104_01', 'hv104_02', 'hv104_03', 'hv104_04', 'hv104_05',
'hv104_06', 'hv104_07', 'hv104_08', 'hv104_09', 'hv104_10', 'hv104_11',
'hv104_12', 'hv104_13', 'hv104_14', 'hv104_15', 'hv104_16', 'hv104_17',
'hv104_18', 'hv104_19', 'hv104_20', 'hv104_21', 'hv104_22', 'hv104_23',
'hv104_24', 'hv105_01', 'hv105_02', 'hv105_03', 'hv105_04', 'hv105_05',
'hv105_06', 'hv105_07', 'hv105_08', 'hv105_09', 'hv105_10', 'hv105_11',
'hv105_12', 'hv105_13', 'hv105_14', 'hv105_15', 'hv105_16', 'hv105_17',
'hv105_18', 'hv105_19', 'hv105_20', 'hv105_21', 'hv105_22', 'hv105_23',
'hv105_24', 'hv117_01', 'hv117_02', 'hv117_03', 'hv117_04', 'hv117_05',
'hv117_06', 'hv117_07', 'hv117_08', 'hv117_09', 'hv117_10', 'hv117_11',
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'hml32d_15', 'hml32d_16', 'hml32d_17', 'hml32d_18', 'hml32d_19', 'hml32d_20',
'hml32d_21', 'hml32d_22', 'hml32d_23', 'hml32d_24', 'hml32e_01', 'hml32e_02',
'hml32e_03', 'hml32e_04', 'hml32e_05', 'hml32e_06', 'hml32e_07', 'hml32e_08',
'hml32e_09', 'hml32e_10', 'hml32e_11', 'hml32e_12', 'hml32e_13', 'hml32e_14',
'hml32e_15', 'hml32e_16', 'hml32e_17', 'hml32e_18', 'hml32e_19', 'hml32e_20',
'hml32e_21', 'hml32e_22', 'hml32e_23', 'hml32e_24', 'hml32f_01', 'hml32f_02',
'hml32f_03', 'hml32f_04', 'hml32f_05', 'hml32f_06', 'hml32f_07', 'hml32f_08',
'hml32f_09', 'hml32f_10', 'hml32f_11', 'hml32f_12', 'hml32f_13', 'hml32f_14',
'hml32f_15', 'hml32f_16', 'hml32f_17', 'hml32f_18', 'hml32f_19', 'hml32f_20',
'hml32f_21', 'hml32f_22', 'hml32f_23', 'hml32f_24', 'hml32g_01', 'hml32g_02',
'hml32g_03', 'hml32g_04', 'hml32g_05', 'hml32g_06', 'hml32g_07', 'hml32g_08',
'hml32g_09', 'hml32g_10', 'hml32g_11', 'hml32g_12', 'hml32g_13', 'hml32g_14',
'hml32g_15', 'hml32g_16', 'hml32g_17', 'hml32g_18', 'hml32g_19', 'hml32g_20',
'hml32g_21', 'hml32g_22', 'hml32g_23', 'hml32g_24', 'hml33_01', 'hml33_02',
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'hml34_09', 'hml34_10', 'hml34_11', 'hml34_12', 'hml34_13', 'hml34_14',
'hml34_15', 'hml34_16', 'hml34_17', 'hml34_18', 'hml34_19', 'hml34_20',
'hml34_21', 'hml34_22', 'hml34_23', 'hml34_24', 'hml35_01', 'hml35_02',
'hml35_03', 'hml35_04', 'hml35_05', 'hml35_06', 'hml35_07', 'hml35_08',
'hml35_09', 'hml35_10', 'hml35_11', 'hml35_12', 'hml35_13', 'hml35_14',
'hml35_15', 'hml35_16', 'hml35_17', 'hml35_18', 'hml35_19', 'hml35_20',
'hml35_21', 'hml35_22', 'hml35_23', 'hml35_24', 'hml36_01', 'hml36_02',
'hml36_03', 'hml36_04', 'hml36_05', 'hml36_06', 'hml36_07', 'hml36_08',
'hml36_09', 'hml36_10', 'hml36_11', 'hml36_12', 'hml36_13', 'hml36_14',
'hml36_15', 'hml36_16', 'hml36_17', 'hml36_18', 'hml36_19', 'hml36_20',
'hml36_21', 'hml36_22', 'hml36_23', 'hml36_24', 'sml16a_01', 'sml16a_02',
'sml16a_03', 'sml16a_04', 'sml16a_05', 'sml16a_06', 'sml16a_07', 'sml16a_08',
'sml16a_09', 'sml16a_10', 'sml16a_11', 'sml16a_12', 'sml16a_13', 'sml16a_14',
'sml16a_15', 'sml16a_16', 'sml16a_17', 'sml16a_18', 'sml16a_19', 'sml16a_20',
'sml16a_21', 'sml16a_22', 'sml16a_23', 'sml16a_24']

```

Renamed the following columns:

```

hv001 -> cluster
hv227 -> net_use
hv012 -> hh_size
hmlidx_1 -> net_idx1
hmlidx_2 -> net_idx2
hmlidx_3 -> net_idx3
hmlidx_4 -> net_idx4
hmlidx_5 -> net_idx5
hmlidx_6 -> net_idx6
hmlidx_7 -> net_idx7

```

Warning: Could not rename the following columns (not found in selection):

```

hv001 -> cluster
hv227 -> net_use
hv012 -> hh_size
hmlidx_1 -> net_idx1
hmlidx_2 -> net_idx2
hmlidx_3 -> net_idx3
hmlidx_4 -> net_idx4
hmlidx_5 -> net_idx5
hmlidx_6 -> net_idx6
hmlidx_7 -> net_idx7

```

First 5 rows of the extracted dataframe:

	cluster	net_use	hh_size	net_idx1	net_idx2	net_idx3	net_idx4	net_idx5	\
0	2	yes	8	1.0	2.0	3.0	4.0	NaN	
1	2	no	7	NaN	NaN	NaN	NaN	NaN	
2	2	yes	6	1.0	2.0	3.0	NaN	NaN	
3	2	yes	1	1.0	NaN	NaN	NaN	NaN	
4	2	yes	4	1.0	NaN	NaN	NaN	NaN	

	net_idx6	net_idx7
0	NaN	NaN
1	NaN	NaN
2	NaN	NaN
3	NaN	NaN
4	NaN	NaN

Successfully extracted 10 columns to exports/household_extr_2020.csv

```

[7]: # Configuration cells - modify these values
input_file = "data/household_2015.csv" # Replace with your actual input file
      ↪path
output_file = "exports/household_extr_2015.csv" # Replace with your desired
      ↪output file path

```

```

# Replace with your identified column names
columns_to_keep = [
    "hv001",
    "hv227",
    "hv012",
    "hmlidx_1",
    "hmlidx_2",
    "hmlidx_3",
    "hmlidx_4",
    "hmlidx_5",
    "hmlidx_6",
    "hmlidx_7"
]

column_rename_dict = {
    "hv001": "cluster",      # Cluster number
    "hv227": "net_use",     # Whether household member slept under ITN last
    ↪night
    "hv012": "hh_size",     # Total number of usual household members
    "hmlidx_1": "netidx_1", # ITN usage index 1
    "hmlidx_2": "netidx_2", # ITN usage index 2
    "hmlidx_3": "netidx_3", # ITN usage index 3
    "hmlidx_4": "netidx_4", # ITN usage index 4
    "hmlidx_5": "netidx_5", # ITN usage index 5
    "hmlidx_6": "netidx_6", # ITN usage index 6
    "hmlidx_7": "netidx_7"  # ITN usage index 7
}

# Execute the function in Jupyter
if not os.path.exists(input_file):
    print(f"Error: Input file '{input_file}' not found.")
else:
    result_df = extract_columns(input_file, output_file, columns_to_keep,
    ↪column_rename_dict)

```

C:\Users\roywaswa\AppData\Local\Temp\ipykernel_28928\2086272929.py:21:

DtypeWarning: Columns (56,88,98,136,139,166,181,182,183,184,185,186,187,188,189,190,191,192,195,196,197,198,199,200,238,239,240,241,242,243,244,245,246,247,248,249,258,259,260,261,262,263,264,265,266,267,268,269,278,279,280,281,282,283,284,285,286,287,288,289,298,299,300,301,302,303,304,305,306,307,308,309,310,311,312,313,314,315,316,317,318,319,320,321,558,559,560,561,562,563,564,565,566,567,568,569,598,599,600,601,602,603,604,605,606,607,608,609,828,829,830,831,832,833,834,835,836,1020,1021,1022,1023,1024,1025,1026,1027,1028,1080,1081,1082,1083,1084,1085,1086,1087,1088,1089,1136,1137,1138,1139,1140,1141,1142,1143,1144,1145,1146,1147,1148,1149,1150,1151,1164,1165,1166,1167,1168,1169,1170,1171,1172,1173,1177,1178,1179,1180,1181,1182,1185,1192,1193,1194,1195,1196,1197,1198,1199,1200,1201,1202,1203,1204,1205,1206,1207,1208,1209,1210,1211,1212,1213,1214,1215,1216,1217,1218,1219,1220,1221,1222,1223,1224,1225,1226,1227,1228,1229,1248,1249,1250,1251,1252,1253,1254,1255,1256,1257,1258,1259,1260,1261,1262,1263,1264,1265,1266,1267,1268,1269,1270,1271,1272,1273,1274,1275,1276,1277,1278,1279,1280,1281,1282,1283,1284,1285,1286,1287,1288,1289,1290,1291,1292,1293,1294,1295,1296,1297,1298,1299,1300,1301,1302,1303,1304,1305,1306,1307,1308,1309,1310,1311,1312,1313,1314,1315,1316,1317,1318,1319,1320,1321,1322,1323,1324,1325,1326,1327,1328,1329,1330,1331,1332,1333,1334,1335,1336,1337,1338,1339,1340,1341,1342,1343,1344,1345,1346,1347,1348,1349,1350,1351,1352,1353,1354,1355,1356,1357,1358,1359,1360,1361,1362,1363,1364,1365,1366,1367,1368,1369,1370,1371,1372,1373,1374,1375,1376,1377,1378,1379,1380,1381,1382,1383,1384,1385,1386,1387,1388,1389,1390,1391,1392,1393,1394,1395,1396,1397,1398,1399,1400,1401,1402,1403,1404,1405,1406,1407,1408,1409,1410,1411,1412,1413,1414,1415,1416,1417,1418,1419,1420,1421,1422,1423,1424,1425,1426,1427,1428,1429,1430,1431,1432,1433,1434,1435,1436,1437,1438,1439,1440,1441,1442,1443,1444,1445,1446,1447,1448,1449,1450,1451,1452,1453,1454,1455,1456,1457,1458,1459,1460,1461,1462,1463,1464,1465,1466,1467,1468,1469,1470,1471,1472,1473,1474,1475,1476,1477,1478,1479,1480,1481,1482,1483,1484,1485,1486,1487,1488,1489,1490,1491,1492,1493,1494,1495,1496,1497,1498,1499,1500,1501,1502,1503,1504,1505,1506,1507,1508,1509,1510,1511,1512,1513,1514,1515,1516,1517,1518,1519,1520,1521,1522,1523,1524,1525,1526,1527,1528,1529,1530,1531,1532,1533,1534,1535,1536,1537,1538,1539,1540,1541,1542,1543,1544,1545,1546,1547,1548,1549,1550,1551,1552,1553,1554,1555,1556,1557,1558,1559,1560,1561,1562,1563,1564,1565,1566,1567,1568,1569,1570,1571,1572,1573,1574,1575,1576,1577,1578,1579,1580,1581,1582,1583,1584,1585,1586,1587,1588,1589,1590,1591,1592,1593,1594,1595,1596,1597,1598,1599,1600,1601,1602,1603,1604,1605,1606,1607,1608,1609,1610,1611,1612,1613,1614,1615,1616,1617,1618,1619,1620,1621,1622,1623,1624,1625,1626,1627,1628,1629,1630,1631,1632,1633,1634,1635,1636,1637,1638,1639,1640,1641,1642,1643,1644,1645,1646,1647,1648,1649,1650,1651,1652,1653,1654,1655,1656,1657,1658,1659,1660,1661,1662,1663,1664,1665,1666,1667,1668,1669,1670,1671,1672,1673,1674,1675,1676,1677,1678,1679,1680,1681,1682,1683,1684,1685,1686,1687,1688,1689,1690,1691,1692,1693,1694,1695,1696,1697,1698,1699,1700,1701,1702,1703,1704,1705,1706,1707,1708,1709,1710,1711,1712,1713,1714,1715,1716,1717,1718,1719,1720,1721,1722,1723,1724,1725,1726,1727,1728,1729,1730,1731,1732,1733,1734,1735,1736,1737,1738,1739,1740,1741,1742,1743,1744,1745,1746,1747,1748,1749,1750,1751,1752,1753,1754,1755,1756,1757,1758,1759,1760,1761,1762,1763,1764,1765,1766,1767,1768,1769,1770,1771,1772,1773,1774,1775,1776,1777,1778,1779,1780,1781,1782,1783,1784,1785,1786,1787,1788,1789,1790,1791,1792,1793,1794,1795,1796,1797,1798,1799,1800,1801,1802,1803,1804,1805,1806,1807,1808,1809,1810,1811,1812,1813,1814,1815,1816,1817,1818,1819,1820,1821,1822,1823,1824,1825,1826,1827,1828,1829,1830,1831,1832,1833,1834,1835,1836,1837,1838,1839,1840,1841,1842,1843,1844,1845,1846,1847,1848,1849,1850,1851,1852,1853,1854,1855,1856,1857,1858,1859,1860,1861,1862,1863,1864,1865,1866,1867,1868,1869,1870,1871,1872,1873,1874,1875,1876,1877,1878,1879,1880,1881,1882,1883,1884,1885,1886,1887,1888,1889,1890,1891,1892,1893,1894,1895,1896,1897,1898,1899,1900,1901,1902,1903,1904,1905,1906,1907,1908,1909,1910,1911,1912,1913,1914,1915,1916,1917,1918,1919,1920,1921,1922,1923,1924,1925,1926,1927,1928,1929,1930,1931,1932,1933,1934,1935,1936,1937,1938,1939,1940,1941,1942,1943,1944,1945,1946,1947,1948,1949,1950,1951,1952,1953,1954,1955,1956,1957,1958,1959,1960,1961,1962,1963,1964,1965,1966,1967,1968,1969,1970,1971,1972,1973,1974,1975,1976,1977,1978,1979,1980,1981,1982,1983,1984,1985,1986,1987,1988,1989,1990,1991,1992,1993,1994,1995,1996,1997,1998,1999,2000,2001,2002,2003,2004,2005,2006,2007,2008,2009,2010,2011,2012,2013,2014,2015,2016,2017,2018,2019,2020,2021,2022,2023,2024,2025,2026,2027,2028,2029,2030,2031,2032,2033,2034,2035,2036,2037,2038,2039,2040,2041,2042,2043,2044,2045,2046,2047,2048,2049,2050,2051,2052,2053,2054,2055,2056,2057,2058,2059,2060,2061,2062,2063,2064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First 5 rows of the original dataframe:

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[5 rows x 2362 columns]

All columns in the CSV file:

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'shb118a_01', 'shb118a_02', 'shb118a_03', 'shb118a_04', 'shb118a_05',
'shb118a_06', 'shb118a_07', 'shb118a_08', 'shb118a_09', 'shb118a_10',
'shb118a_11', 'shb118a_12', 'shb118a_13', 'shb118a_14', 'shb118a_15',

'shb118a_16', 'shb118a_17', 'shb118a_18', 'shb118a_19', 'shb118a_20',
'shb118b_01', 'shb118b_02', 'shb118b_03', 'shb118b_04', 'shb118b_05',
'shb118b_06', 'shb118b_07', 'shb118b_08', 'shb118b_09', 'shb118b_10',
'shb118b_11', 'shb118b_12', 'shb118b_13', 'shb118b_14', 'shb118b_15',
'shb118b_16', 'shb118b_17', 'shb118b_18', 'shb118b_19', 'shb118b_20',
'shb118c_01', 'shb118c_02', 'shb118c_03', 'shb118c_04', 'shb118c_05',
'shb118c_06', 'shb118c_07', 'shb118c_08', 'shb118c_09', 'shb118c_10',
'shb118c_11', 'shb118c_12', 'shb118c_13', 'shb118c_14', 'shb118c_15',
'shb118c_16', 'shb118c_17', 'shb118c_18', 'shb118c_19', 'shb118c_20',
'shb118d_01', 'shb118d_02', 'shb118d_03', 'shb118d_04', 'shb118d_05',
'shb118d_06', 'shb118d_07', 'shb118d_08', 'shb118d_09', 'shb118d_10',
'shb118d_11', 'shb118d_12', 'shb118d_13', 'shb118d_14', 'shb118d_15',
'shb118d_16', 'shb118d_17', 'shb118d_18', 'shb118d_19', 'shb118d_20',
'shb118e_01', 'shb118e_02', 'shb118e_03', 'shb118e_04', 'shb118e_05',
'shb118e_06', 'shb118e_07', 'shb118e_08', 'shb118e_09', 'shb118e_10',
'shb118e_11', 'shb118e_12', 'shb118e_13', 'shb118e_14', 'shb118e_15',
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'shb118f_01', 'shb118f_02', 'shb118f_03', 'shb118f_04', 'shb118f_05',
'shb118f_06', 'shb118f_07', 'shb118f_08', 'shb118f_09', 'shb118f_10',
'shb118f_11', 'shb118f_12', 'shb118f_13', 'shb118f_14', 'shb118f_15',
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'shb118g_06', 'shb118g_07', 'shb118g_08', 'shb118g_09', 'shb118g_10',
'shb118g_11', 'shb118g_12', 'shb118g_13', 'shb118g_14', 'shb118g_15',
'shb118g_16', 'shb118g_17', 'shb118g_18', 'shb118g_19', 'shb118g_20',
'shb118h_01', 'shb118h_02', 'shb118h_03', 'shb118h_04', 'shb118h_05',
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'shb121_13', 'shb121_14', 'shb121_15', 'shb121_16', 'shb121_17', 'shb121_18',
'shb121_19', 'shb121_20', 'shb125_01', 'shb125_02', 'shb125_03', 'shb125_04',
'shb125_05', 'shb125_06', 'shb125_07', 'shb125_08', 'shb125_09', 'shb125_10',
'shb125_11', 'shb125_12', 'shb125_13', 'shb125_14', 'shb125_15', 'shb125_16',
'shb125_17', 'shb125_18', 'shb125_19', 'shb125_20', 'shml16a_01', 'shml16a_02',
'shml16a_03', 'shml16a_04', 'shml16a_05', 'shml16a_06', 'shml16a_07',
'shml16a_08', 'shml16a_09', 'shml16a_10', 'shml16a_11', 'shml16a_12',
'shml16a_13', 'shml16a_14', 'shml16a_15', 'shml16a_16', 'shml16a_17',
'shml16a_18', 'shml16a_19', 'shml16a_20', 'shb108t_01', 'shb108t_02',
'shb108t_03', 'shb108t_04', 'shb108t_05', 'shb108t_06', 'shb108t_07',
'shb108t_08', 'shb108t_09', 'shb108t_10', 'shb108t_11', 'shb108t_12',
'shb108t_13', 'shb108t_14', 'shb108t_15', 'shb108t_16', 'shb108t_17',
'shb108t_18', 'shb108t_19', 'shb108t_20', 'shb110t_01', 'shb110t_02',
'shb110t_03', 'shb110t_04', 'shb110t_05', 'shb110t_06', 'shb110t_07',
'shb110t_08', 'shb110t_09', 'shb110t_10', 'shb110t_11', 'shb110t_12',
'shb110t_13', 'shb110t_14', 'shb110t_15', 'shb110t_16', 'shb110t_17',
'shb110t_18', 'shb110t_19', 'shb110t_20']

Renamed the following columns:

```
hv001 -> cluster
hv227 -> net_use
hv012 -> hh_size
hmlidx_1 -> netidx_1
hmlidx_2 -> netidx_2
hmlidx_3 -> netidx_3
hmlidx_4 -> netidx_4
hmlidx_5 -> netidx_5
hmlidx_6 -> netidx_6
hmlidx_7 -> netidx_7
```

Warning: Could not rename the following columns (not found in selection):

```
hv001 -> cluster
hv227 -> net_use
hv012 -> hh_size
hmlidx_1 -> netidx_1
hmlidx_2 -> netidx_2
hmlidx_3 -> netidx_3
hmlidx_4 -> netidx_4
hmlidx_5 -> netidx_5
hmlidx_6 -> netidx_6
hmlidx_7 -> netidx_7
```

First 5 rows of the extracted dataframe:

	cluster	net_use	hh_size	netidx_1	netidx_2	netidx_3	netidx_4	netidx_5	\
0	1	yes	1	1.0	NaN	NaN	NaN	NaN	
1	1	no	1	NaN	NaN	NaN	NaN	NaN	
2	1	yes	1	1.0	NaN	NaN	NaN	NaN	
3	1	yes	3	1.0	NaN	NaN	NaN	NaN	
4	1	no	1	NaN	NaN	NaN	NaN	NaN	

	netidx_6	netidx_7
0	NaN	NaN
1	NaN	NaN
2	NaN	NaN
3	NaN	NaN
4	NaN	NaN

Successfully extracted 10 columns to exports/household_extr_2015.csv

0.2 Data Aggregation

```
[8]: import pandas as pd
import numpy as np
from typing import Dict, List, Union, Callable

def aggregate_by_cluster(
    df: pd.DataFrame,
    cluster_column: str,
    category_columns: List[str] = None,
    aggregation_methods: Dict[str, Union[str, List[str], Callable]] = None
) -> pd.DataFrame:
    """
    Aggregate a dataframe by cluster with separate count columns for
    ↪categorical variables.

    Parameters:
    -----
    df : pandas DataFrame
        The input dataframe to be aggregated
    cluster_column : str
        The column name containing cluster assignments
    category_columns : list, optional
        List of categorical columns to create separate count columns for each
    ↪unique value
    aggregation_methods : dict, optional
        Dictionary mapping column names to aggregation methods
        If a column is not specified, default aggregations will be applied

    Returns:
    -----
    pandas DataFrame
        Aggregated dataframe with one row per cluster and separate count
    ↪columns for categories
    """
    if aggregation_methods is None:
        aggregation_methods = {}

    if category_columns is None:
        category_columns = []

    # Make a copy of the dataframe to avoid modifying the original
    data = df.copy()

    # Create result dataframe starting with cluster counts
    result = pd.DataFrame(data[cluster_column].value_counts().sort_index())
    result.columns = ['total_count']
```

```

# Process each categorical column to create separate count columns
for col in category_columns:
    if col in data.columns and col != cluster_column:
        # Get unique values in this column
        unique_values = data[col].dropna().unique()

        # For each unique value, create a count column
        for val in unique_values:
            # Create a new column with count of this value in each cluster
            count_series = data[data[col] == val].groupby(cluster_column).
↳size()

            result[f"{col}_{val}"] = count_series

# Handle the remaining columns with the standard aggregation function
# Remove category columns from aggregation_methods
remaining_agg_methods = {k: v for k, v in aggregation_methods.items()
                        if k not in category_columns and k !=
↳cluster_column}

# Identify remaining columns
remaining_cols = [col for col in data.columns
                 if col not in category_columns and col != cluster_column]

if remaining_cols:
    # Create a subset dataframe with only the remaining columns
    subset_df = data[[cluster_column] + remaining_cols].copy()

    # Apply standard aggregation to remaining columns
    standard_aggs = {}

    # Identify numeric and categorical columns
    numeric_cols = subset_df.select_dtypes(include=['number']).columns.
↳tolist()
    cat_cols = subset_df.select_dtypes(include=['object', 'category']).
↳columns.tolist()

    if cluster_column in numeric_cols:
        numeric_cols.remove(cluster_column)
    if cluster_column in cat_cols:
        cat_cols.remove(cluster_column)

    # Apply user-specified aggregation methods
    for col, agg_method in remaining_agg_methods.items():
        if col in subset_df.columns:
            standard_aggs[col] = agg_method

```

```

    # Apply default aggregations for numeric columns not specified
    for col in numeric_cols:
        if col not in standard_aggs:
            standard_aggs[col] = ['mean', 'median', 'min', 'max']

    # Apply default aggregations for remaining categorical columns
    for col in cat_cols:
        if col not in standard_aggs:
            standard_aggs[col] = [
                ('mode', lambda x: x.mode()[0] if not x.mode().empty else
↳ np.nan),
                ('unique_count', 'nunique')
            ]

    if standard_aggs:
        # Perform standard aggregation
        standard_result = subset_df.groupby(cluster_column).
↳ agg(standard_aggs)

        # Flatten the column names if multi-level
        if isinstance(standard_result.columns, pd.MultiIndex):
            standard_result.columns = [f"{col[0]}_{col[1]}" if
↳ isinstance(col, tuple) else col
            for col in standard_result.columns]

        # Join with the result dataframe
        result = result.join(standard_result)

    # Fill NaN values with 0 for count columns
    count_cols = [col for col in result.columns if col != 'total_count' and not
↳ col.startswith(('age_', 'stratum_', 'region_'))]
    result[count_cols] = result[count_cols].fillna(0).astype(int)

    return result

```

```

[9]: def aggregate_dataset(
    df,
    cluster_column='cluster',
    category_columns=None,
    numeric_columns=None,
    simple_agg_columns=None,
    custom_agg_methods=None,
    file_format='csv',
    **kwargs
):
    """

```

*Reads a dataset file and performs cluster-based aggregation with
↳ customizable settings.*

Parameters:

file_path : str

Path to the data file

cluster_column : str, default='cluster'

The column name containing cluster assignments

category_columns : list or None, default=None

*List of categorical columns to create separate count columns for each
↳ unique value*

numeric_columns : list or None, default=None

List of numeric columns to apply numeric aggregations to

simple_agg_columns : dict or None, default=None

*Dictionary mapping column names to simple aggregation method (e.g.,
↳ {'region': 'first'})*

custom_agg_methods : dict or None, default=None

Dictionary mapping column names to custom aggregation methods

file_format : str, default='csv'

Format of input file ('csv', 'excel', etc.)

***kwargs :*

*Additional arguments to pass to the file reading function (e.g.,
↳ sep=',', encoding='utf-8')*

Returns:

pandas DataFrame

Aggregated dataframe with one row per cluster

"""

```
import pandas as pd
```

```
import numpy as np
```

```
# Set defaults for optional parameters
```

```
if category_columns is None:
```

```
    category_columns = []
```

```
if numeric_columns is None:
```

```
    numeric_columns = []
```

```
if simple_agg_columns is None:
```

```
    simple_agg_columns = {}
```

```
if custom_agg_methods is None:
```

```
    custom_agg_methods = {}
```

```
# Convert numeric columns if specified
```

```
for col in numeric_columns:
```

```
    if col in df.columns:
```

```
        df[col] = pd.to_numeric(df[col], errors='coerce')
```

```

# Prepare aggregation methods
agg_methods = {}

# Add simple aggregations (like 'first', 'last', etc.)
for col, method in simple_agg_columns.items():
    if col in df.columns and col != cluster_column:
        agg_methods[col] = method

# Add custom aggregation methods
for col, method in custom_agg_methods.items():
    if col in df.columns and col != cluster_column:
        agg_methods[col] = method

# Perform the aggregation
result = aggregate_by_cluster(df, cluster_column, category_columns,
↪agg_methods)
return result

```

```

[10]: result = aggregate_dataset(
    pd.read_csv('exports/children_extr_2015.csv'),
    cluster_column='cluster',
    category_columns=['treat_src', 'residence', 'alive', 'fever_rec'],
    numeric_columns=['age_mo'],
    simple_agg_columns={
        'stratum': 'first',
        'region': 'first'
    },
    custom_agg_methods={
        'age_mo': ['mean', 'min', 'max', 'count']
    }
)

result_foc = [
    'total_count',
    'treat_src_yes',
    'residence_urban',
    'residence_rural',
    'alive_yes',
    'fever_rec_only treated nets'
]

children_2015 = result[result_foc].reset_index()

```

```

[11]: result = aggregate_dataset(
    pd.read_csv('exports/children_extr_2020.csv'),
    cluster_column='cluster',

```

```

        category_columns=['treat_src', 'residence', 'alive', 'fever_rec'],
        numeric_columns=['age_mo'],
        simple_agg_columns={
            'stratum': 'first',
            'region': 'first'
        },
        custom_agg_methods={
            'age_mo': ['mean', 'min', 'max', 'count']
        }
    )

result_foc = [
    'total_count',
    'treat_src_yes',
    'residence_urban',
    'residence_rural',
    'alive_yes',
    'fever_rec_only treated nets'
]

children_2020 = result[result_foc].reset_index()

```

```

[12]: df = pd.read_csv('exports/household_extr_2015.csv')
col_max = df.loc[:, 'netidx_1': 'netidx_7'].columns.to_list()
df['nets'] = df[col_max].max(axis=1).fillna(0)
df = df.drop(df.loc[:, 'netidx_1': 'netidx_7'].columns.to_list(), axis=1)

result = aggregate_dataset(
    df,
    cluster_column='cluster',
    category_columns=['net_use'],
    numeric_columns=['nets', 'hh_size'],
    # simple_agg_columns={},
    custom_agg_methods={
        'nets': ['mean', 'min', 'max', 'count', 'sum'],
        'hh_size': ['mean', 'min', 'max', 'count', 'sum']
    }
)

household_2015 = result[
    ['total_count', 'net_use_yes', 'nets_sum', 'hh_size_mean', 'hh_size_sum']
].reset_index()
# household_2015

```

```

[13]: df = pd.read_csv('exports/household_extr_2020.csv')
col_max = df.loc[:, 'net_idx1': 'net_idx7'].columns.to_list()
df['nets'] = df[col_max].max(axis=1).fillna(0)
df = df.drop(df.loc[:, 'net_idx1': 'net_idx7'].columns.to_list(), axis=1)

```



```

result = aggregate_dataset(
    df,
    cluster_column='cluster',
    category_columns=['net_use'],
    numeric_columns=['nets', 'hh_size'],
    # simple_agg_columns={},
    custom_agg_methods={
        'nets': ['mean', 'min', 'max', 'count', 'sum'],
        'hh_size': ['mean', 'min', 'max', 'count', 'sum']
    }
)
household_2020 = result[
    ['total_count', 'net_use_yes', 'nets_sum', 'hh_size_mean', 'hh_size_sum']
].reset_index()
# household_2020

```

```

[14]: df = pd.read_csv('exports/household_member-extr_2015.csv')

result = aggregate_dataset(
    df,
    cluster_column='cluster',
    category_columns=['residence', 'net_used', 'net_owned']
)
result
household_member_2015 = result[
    ['total_count',
    ↪ 'residence_urban', 'net_used_yes', 'region_mode', 'net_owned_only treated (itn)',
    ↪ 'nets', 'net_owned_did not sleep under a net']
].reset_index()
# household_member_2015

```

```

[15]: df = pd.read_csv('exports/household_member-extr_2020.csv')

result = aggregate_dataset(
    df.drop(['county'], axis=1),
    cluster_column='cluster',
    category_columns=['residence', 'net_owned'],

)
result
household_member_2020 = result[
    ['total_count', 'residence_urban', 'region_mode', 'net_owned_only treated',
    ↪ (itn) nets', 'net_owned_did not sleep under a net']
].reset_index()
# household_member_2020

```

0.3 Data Joining

0.3.1 2015 data

The 2015 data is joined to the spatial points of data collection according to cluster, then to the spatial covariates according to the cluster as well.

```
[17]: # Importing the shapefile for 2015
import geopandas as gpd

path_to_shp = './data/shp_2015/KEGE7AFL.shp'
dhs15_points_shp = gpd.read_file(path_to_shp)
dhs15_points_shp.columns
dhs15_points_shp = dhs15_points_shp.drop(dhs15_points_shp.loc[:, 'CCFIPS':
↳ 'SOURCE'].columns.to_list(), axis=1)

[18]: dhs15_points_shp
meg_15_ch = pd.merge(dhs15_points_shp, children_2015, how='inner',
↳ right_on='cluster', left_on='DHSCLUST', suffixes=('_sh', '_ch'))
meg_15_ch_hh = pd.merge(meg_15_ch, household_2015, how='inner',
↳ right_on='cluster', left_on='DHSCLUST', suffixes=('_shch', '_hh'))
meg_15_ch_hh_hhm = pd.merge(meg_15_ch_hh, household_member_2015, how='inner',
↳ right_on='cluster', left_on='DHSCLUST', suffixes=('_shchhh', '_hhm'))
meg_15_ch_hh_hhm.to_csv('meg15.csv')

[19]: def isolate_columns(columns, year_suffix="_15"):
    """
    Isolates columns ending with the specified suffix and constant columns
↳ without numerical suffixes,
    excluding temperature columns from constants

    Args:
        columns (list): List of column names
        year_suffix (str): Suffix to filter for (default: "_15")

    Returns:
        tuple: (columns_with_suffix, constant_columns, combined_columns)
    """
    import re

    # Find columns ending with the specified suffix
    # Handle both formats like "_15" and just "15"
    columns_with_suffix = [col for col in columns if col.endswith(year_suffix)
↳ or
    (year_suffix.startswith('_') and col.
↳ endswith(year_suffix[1:]))]

    # Find constant columns without numerical suffix
```

```

# Pattern to match columns ending with numbers
pattern = re.compile(r'\d+$')

# Columns that don't end with numbers
constant_columns = [col for col in columns if not pattern.search(col)]

# Remove temperature columns from the constant columns
constant_columns = [col for col in constant_columns
                    if not col.startswith('Temperature_')]

# Combined list of both types
combined_columns = columns_with_suffix + constant_columns

return columns_with_suffix, constant_columns, combined_columns

```

```

[22]: path_to_cov = './data/geospatial_covariates_2015.csv'
df = pd.read_csv(path_to_cov)
x,y,select_cols = isolate_columns(df.columns.to_list())
cov_15 = df[select_cols]

```

```

[23]: mal_2015 = pd.merge(meg_15_ch_hh_hhm, cov_15, 'inner',on=['DHSCLUST','DHSID'],
    ↪suffixes=('_hhm','_cov'))
mal_2015.to_csv('malaria_15.csv')
gpd.GeoDataFrame(mal_2015,geometry='geometry').to_file('shp/malaria_15.shp',
    ↪driver='ESRI Shapefile')

```

C:\Users\roywaswa\AppData\Local\Temp\ipykernel_28928\2880136839.py:3:
 UserWarning: Column names longer than 10 characters will be truncated when saved
 to ESRI Shapefile.

```

gpd.GeoDataFrame(mal_2015,geometry='geometry').to_file('shp/malaria_15.shp',
driver='ESRI Shapefile')

```

c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-

packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
 'DHSYEAR_hhm' to 'DHSYEAR_hh'

```

ogr_write(

```

c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-

packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
 'cluster_shch' to 'cluster_sh'

```

ogr_write(

```

c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-

packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
 'total_count_shch' to 'total_coun'

```

ogr_write(

```

c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-

packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
 'treat_src_yes' to 'treat_src_'

```

ogr_write(

```

c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-

```

packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'residence_urban_shchhh' to 'residence_'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'residence_rural' to 'residenc_1'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'fever_rec_only treated nets' to 'fever_rec_'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'total_count_hh' to 'total_co_1'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'net_use_yes' to 'net_use_ye'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'hh_size_mean' to 'hh_size_me'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'hh_size_sum' to 'hh_size_su'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'total_count' to 'total_co_2'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'residence_urban_hhm' to 'residenc_2'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'net_used_yes' to 'net_used_y'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'region_mode' to 'region_mod'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'net_owned_only treated (itn) nets' to 'net_owned_'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-

```

```

packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'net_owned_did not sleep under a net' to 'net_owne_1'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'Travel_Times_2015' to 'Travel_Tim'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'GPS_Dataset' to 'GPS_Dataset'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'DHSYEAR_cov' to 'DHSYEAR_co'
    ogr_write(

```

0.3.2 2020 data

The 2020 data is joined to the spatial points of data collection according to cluster, then to the spatial covariates according to the cluster as well.

```

[24]: path_to_shp = './data/shp_2020/KEGE81FL.shp'
      dhs20_points_shp = gpd.read_file(path_to_shp)
      dhs20_points_shp.columns
      dhs20_points_shp = dhs20_points_shp.drop(dhs20_points_shp.loc[:, 'CCFIPS':
      ↪ 'SOURCE'].columns.to_list(), axis=1)

[25]: dhs20_points_shp
      meg_20_ch = pd.merge(dhs20_points_shp, children_2020, how='inner',
      ↪ right_on='cluster', left_on='DHSCLUST', suffixes=('_sh', '_ch'))
      meg_20_ch_hh = pd.merge(meg_20_ch, household_2020, how='inner',
      ↪ right_on='cluster', left_on='DHSCLUST', suffixes=('_shch', '_hh'))
      meg_20_ch_hh_hhm = pd.merge(meg_20_ch_hh, household_member_2020, how='inner',
      ↪ right_on='cluster', left_on='DHSCLUST', suffixes=('_shchhh', '_hhm'))
      meg_20_ch_hh_hhm.to_csv('meg20.csv')

[ ]: path_to_cov = './data/geospatial_covariates_2020.csv'
      df = pd.read_csv(path_to_cov)
      x,y,select_cols = isolate_columns(df.columns.to_list(), '_20')
      cov_20 = df[select_cols]
      select_cols

[32]: mal_2020 = pd.merge(meg_20_ch_hh_hhm, cov_20, 'inner', on=['DHSCLUST', 'DHSID'],
      ↪ suffixes=('_hhm', '_cov'))
      mal_2020.to_csv('malaria_20.csv')
      mal_20_gdf = gpd.GeoDataFrame(mal_2020, geometry='geometry')
      mal_20_gdf.to_file('shp/malaria_20.shp', driver='ESRI Shapefile')

```

```

C:\Users\roywaswa\AppData\Local\Temp\ipykernel_28928\1971534842.py:4:
UserWarning: Column names longer than 10 characters will be truncated when saved
to ESRI Shapefile.
    mal_20_gdf.to_file('shp/malaria_20.shp', driver='ESRI Shapefile')
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'DHSYEAR_hhm' to 'DHSYEAR_hh'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'cluster_shch' to 'cluster_sh'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'total_count_shch' to 'total_coun'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'treat_src_yes' to 'treat_src_'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'residence_urban_shchhh' to 'residence_'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'residence_rural' to 'residenc_1'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'fever_rec_only treated nets' to 'fever_rec_'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'total_count_hh' to 'total_co_1'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'net_use_yes' to 'net_use_ye'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'hh_size_mean' to 'hh_size_me'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'hh_size_sum' to 'hh_size_su'
    ogr_write(

```

```

c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'total_count' to 'total_co_2'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'residence_urban_hhm' to 'residenc_2'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'region_mode' to 'region_mod'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'net_owned_only treated (itn) nets' to 'net_owned_'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'net_owned_did not sleep under a net' to 'net_owne_1'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'GPS_Dataset' to 'GPS_Datase'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'DHSYEAR_cov' to 'DHSYEAR_co'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'Global_Human_Footprint' to 'Global_Hum'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'Growing_Season_Length' to 'Growing_Se'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'Livestock_Cattle' to 'Livestock_'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'Livestock_Chickens' to 'Livestoc_1'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/laundered field name:
'Livestock_Goats' to 'Livestoc_2'
    ogr_write(

```

```
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/launched field name:
'Livestock_Pigs' to 'Livestoc_3'
    ogr_write(
c:\Users\roywaswa\AppData\Roaming\mamba\envs\dsa\Lib\site-
packages\pyogrio\raw.py:723: RuntimeWarning: Normalized/launched field name:
'Livestock_Sheep' to 'Livestoc_4'
    ogr_write(
```

```
[28]: gdf = gpd.read_file('shp/malaria_20.shp')
```

```
[29]: import pandas as pd

def rename_duplicate_columns(df):
    # Define mapping for affected columns - all names limited to 10 characters
    rename_map = {
        # Total count columns
        'total_coun': 'tot_cnt1',
        'total_co_1': 'tot_cnt2',
        'total_co_2': 'tot_cnt3',

        # Residence columns
        'residence_': 'res_type',
        'residenc_1': 'res_cat',
        'residenc_2': 'res_regn',

        # Cluster columns
        'cluster': 'cluster', # keep as is
        'cluster_sh': 'clstr_sh',
        'cluster_hh': 'clstr_hh',

        # Net owned columns
        'net_owned_': 'net_own1',
        'net_owne_1': 'net_own2',

        # Livestock columns
        'Livestock_': 'lvstk',
        'Livestoc_1': 'lvstk_t1',
        'Livestoc_2': 'lvstk_t2',
        'Livestoc_3': 'lvstk_t3',
        'Livestoc_4': 'lvstk_t4',

        # Temperature columns
        'DTemp20': 'DTemp20', # keep as is
        'Matemp20': 'MaxTmp20',
        'Meantem20': 'MeanTm20',
        'Mintem20': 'MinTmp20'
```



```

}

# Create a new DataFrame with renamed columns
# Only rename columns that exist in the dataframe
columns_to_rename = {col: rename_map[col] for col in df.columns if col in rename_map}

if columns_to_rename:
    df = df.rename(columns=columns_to_rename)
    print(f"Renamed {len(columns_to_rename)} columns")
else:
    print("No columns were renamed")

return df

# Rename the columns
df_renamed = rename_duplicate_columns(gdf)

```

Renamed 20 columns

```
[30]: df_renamed.to_file('trial.shp')
```

```
[31]: df_renamed
```

```
[31]:
```

	DHSID	DHSCC_hhm	DHSYEAR_hh	DHSCLUST	URBAN_RURA	LATNUM	\
0	KE202000000002	KE	2020.0	2.0	R	-4.200655	
1	KE202000000005	KE	2020.0	5.0	R	-4.027349	
2	KE202000000012	KE	2020.0	12.0	R	-4.510408	
3	KE202000000015	KE	2020.0	15.0	R	-4.492379	
4	KE202000000028	KE	2020.0	28.0	R	-3.183536	
..	...	...	...	...	...	...	
292	KE202000009165	KE	2020.0	9165.0	R	0.754836	
293	KE202000009168	KE	2020.0	9168.0	R	0.847692	
294	KE202000009180	KE	2020.0	9180.0	U	0.579065	
295	KE202000009183	KE	2020.0	9183.0	U	0.623640	
296	KE202000009186	KE	2020.0	9186.0	R	0.468339	

	LONGNUM	ALT_GPS	ALT_DEM	DATUM	...	Global_Hum	Growing_Se	\
0	39.356982	9999.0	201.0	WGS84	...	21.005648	7	
1	39.539536	9999.0	3.0	WGS84	...	29.575237	7	
2	39.165574	9999.0	113.0	WGS84	...	29.566196	8	
3	39.431670	9999.0	17.0	WGS84	...	40.481762	7	
4	40.049758	9999.0	45.0	WGS84	...	44.293491	7	
..	...	...	...	...	...	...		
292	34.466282	9999.0	1382.0	WGS84	...	40.521362	12	
293	34.737363	9999.0	1824.0	WGS84	...	26.806847	10	
294	34.747652	9999.0	1523.0	WGS84	...	42.692383	10	
295	34.802932	9999.0	1556.0	WGS84	...	56.541042	10	

```

296  34.216176    9999.0    1220.0  WGS84  ...   41.190304           12

      Irrigation      lvstk      lvstk_t1      lvstk_t2      lvstk_t3      lvstk_t4  \
0      0.131377    61.196003    97.043762    40.078571    0.051220    12.283060
1      0.014594    13.797853    96.939041    60.921005    0.072594    6.867123
2      0.043812    25.316996    73.750336    13.918099    0.044271    6.713690
3      0.131420    29.142328    221.178894    17.419291    0.069897    6.445805
4      0.116638    27.097311    160.167343    47.589054    0.109326    3.578165
..      ...      ...      ...      ...      ...
292    0.007967    187.912888    444.439026    17.618126    6.940995    21.307005
293    0.022062    67.572197    205.007416    11.273051    1.359833    13.667400
294    0.024951    148.436676    346.842834    15.899426    11.209027    23.006929
295    0.040972    137.295029    607.322510    9.975716    10.395459    14.924463
296    0.033356    65.825012    533.538757    49.584541    23.118988    35.859753

      Nightligh      geometry
0      0.000000  POINT (39.35698 -4.20066)
1      0.141980  POINT (39.53954 -4.02735)
2      0.004836  POINT (39.16557 -4.51041)
3      0.034802  POINT (39.43167 -4.49238)
4      0.083609  POINT (40.04976 -3.18354)
..      ...      ...
292    0.004077  POINT (34.46628 0.75484)
293    0.037909  POINT (34.73736 0.84769)
294    0.028109  POINT (34.74765 0.57907)
295    0.092718  POINT (34.80293 0.62364)
296    0.069459  POINT (34.21618 0.46834)

```

[297 rows x 66 columns]

[]: